

과제 2.

- $V_0 = 20 \text{ m/s}$, $g = 9.81 \text{ m/s}^2$
- 지면 발사 $\rightarrow$ 차지.

$$\textcircled{1} \quad V_{0x} = V_0 \cos \theta_0, \quad V_{0y} = V_0 \sin \theta_0$$

$$a_x = 0, \quad a_y = -g$$

$$\textcircled{2} \quad x = (V_0 \cos \theta_0)t, \quad y = (V_0 \sin \theta_0)t - \frac{1}{2}gt^2$$

$$\textcircled{3} \quad 0 = (V_0 \sin \theta_0)t - \frac{1}{2}gt^2$$

$$0 = t \cdot (V_0 \sin \theta_0 - \frac{1}{2}gt)$$

$$\rightarrow t = 0, \quad \frac{2V_0 \sin \theta_0}{g}$$

$$\textcircled{4} \quad R = (V_0 \cos \theta_0) \times \frac{2V_0 \sin \theta_0}{g} = \frac{2V_0^2 \sin \theta_0 \cos \theta_0}{g}$$

$$= \frac{V_0^2 \sin 2\theta_0}{g}$$

$\theta_0 = 30^\circ \rightarrow \sin 2\theta_0 = 0.9660$	$R = 35.3 \text{ m (a)}$
$\theta_0 = 45^\circ \rightarrow \sin 2\theta_0 = 1.0000$	$R = 40.8 \text{ m (b)}$
$\theta_0 = 60^\circ \rightarrow \sin 2\theta_0 = 0.8660$	$R = 35.3 \text{ m (c)}$

$$\ast \sin \alpha = \sin(\pi - \alpha) \text{ 이므로 } (a) = (c).$$

30°는 수평속도가 크지만 체공시간이 짧고

60°는 체공시간이 길지만 수평속도가 짧다.